

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8499**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600027
		Course Section:	1		

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Name of the candidate. Ninad Sachin Maadhavi

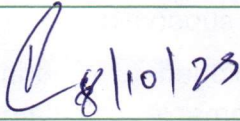
Candidate's UID number:

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Examination class room number:

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Invigilator's signature with date:



(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: CSE - AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday 8/10/2025

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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	2. a)	Input Size = $H \times W \times C$ As the input is 3 dimensional, we will need to add a term $F \times F \times C$ to the filter size As we need to perform standard 2D convolution we will ignore the C term. Size of output image = $(H - F + 1)(W - F + 1)$ Now for each cell of this output image we would have performed $F \times F$ multiple multiplications. So for one filter we perform $(F \times F)(H - F + 1)(W - F + 1)$ multiplications $\therefore$ For K filters $\rightarrow \boxed{K(F \times F)(H - F + 1)(W - F + 1)}$



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2. b) Linearity of a function is defined as
- i) Additivity $\rightarrow f(x_1 + x_2) = f(x_1) + f(x_2)$
 - ii) Homogeneity $\rightarrow f(ax) = a f(x)$

Now if we use linear activation functions in the hidden layer of an ANN then effectively what we are doing is

$$y = \sum_{i=1}^H f(x_i) \quad \text{where } f(x)$$

is the activation function and x_i is input and y is output.

But as we saw in linearity that $f(x_1 + x_2) = f(x_1) + f(x_2)$

$$\therefore y = \sum_{i=1}^H f(x_i) = f\left(\sum_{i=1}^H x_i\right) \quad \text{where}$$

H is the number of Hidden Layers.

So now instead of using multiple Hidden Layers, we are using only 1 due to Linearity.

Now if we pass a set of inputs to a linear undercomplete autoencoder, we will get a single matrix and in the bottleneck layer we will be expressing this matrix in smaller dimensions.

With loss function as MSE, we are basically doing exactly what PCA does. So now we arrange the data on the new axes defined by the dimensional basis and calculate the loss. Very similar to selecting top Eigen values in PCA with a law of Diminishing Returns if we oversample.

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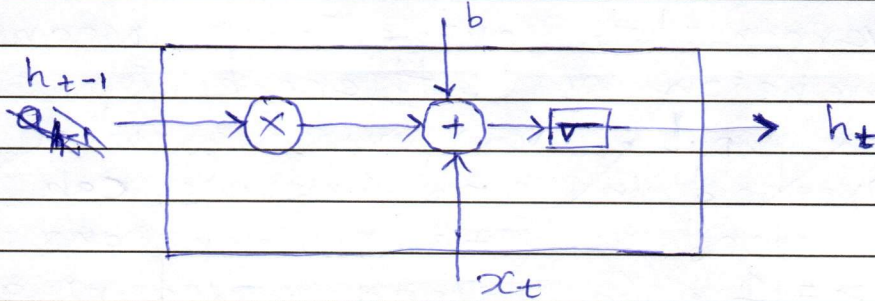
Overcomplete autoencoder means dimension of bottleneck is greater than input. Now anything which can be expressed in n -dimensions can also be expressed in $(n+k)$ -dimensions where $k \geq 1$. So instead of learning any pattern in the data, the model learns a trivial solution of identity mapping where output is constructed exactly like the input.

Regularization penalizes the model for identity mapping. A prominent example is Denoising AutoEncoder which introduced Gaussian Noise in Input and we train the model to predict output without the noise. Sparse Sparse Autoencoders penalize neurons with L1. Even drop-out and early stopping or weight sharing where weights of Decoder are Transpose of Weight Matrix of Encoder which constraints the model.

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3.a)



$$h_t = \nabla (W_1 h_{t-1} + W_2 x_t + b)$$

~~We can take $L = \text{Sum of Squared}$~~
Consider $L = -y \log \hat{y} - (1-y) \log (1-\hat{y})$
This is Binary Cross Entropy loss

$$\frac{\partial L}{\partial h_{t-1}} = \frac{\partial L}{\partial h_t} \cdot \frac{\partial h_t}{\partial h_{t-1}} \quad (\text{By Chain Rule}) \quad (1)$$

$$\frac{\partial L}{\partial h_{t-2}} = \frac{\partial L}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial h_{t-2}}$$

We can write this by the same logic
But now we know $\partial L / \partial h_{t-1}$

$$\therefore \frac{\partial L}{\partial h_{t-2}} = \frac{\partial L}{\partial h_t} \cdot \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial h_{t-2}} \quad (\text{By } (1))$$

Now we can generalize this as :

$$\boxed{\frac{\partial L}{\partial h_{t-k}} = \frac{\partial L}{\partial h_t} \prod_{i=1}^k \frac{\partial h_{t-k+1}}{\partial h_{t-k}}}$$

Equation of Forget Gate in an LSTM
is $f_t = \nabla (W_f [h_{t-1}, x_t] + b_f)$ and

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

$$\frac{\partial C_t}{\partial C_{t-1}} = f_t$$

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Now the problem in RNN was Vanishing Gradient. As we can see in the derivation we keep on multiplying derivatives and each derivative contains Sigmoid Activation function which brings the value between 0 and 1.

So Gradient at $t-k = \text{Gradient at } t \times (\text{Number between } 0-1)^k$

Now this is a very small value so earlier timesteps don't get properly trained.

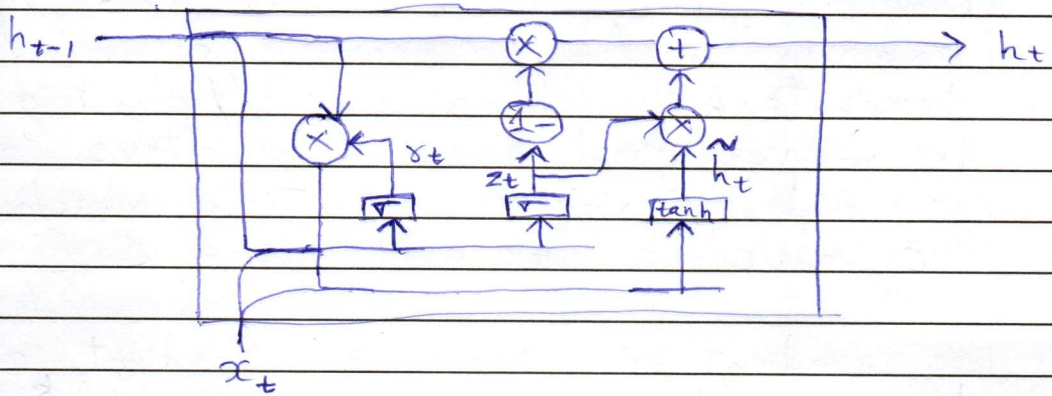
In LSTM however the derivative is only f_t . As long as $f_t \geq 0.8$, a good amount of Gradient flows back to the previous layers which helps in training.

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3.b)



$$\text{Reset Gate } (r_t) = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$$

$$\text{Update Gate } (z_t) = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z)$$

$$\tilde{h}_t = \tanh(W \cdot [h_{t-1} * r_t, x_t] + b)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

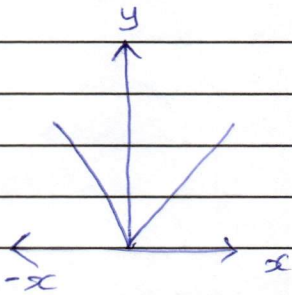
Here we can clearly see that h_t is a combination of both h_{t-1} and $\tilde{h}_t$ and this is controlled by the Update Gate (z_t).

GRU don't have the Cell State and there ~~are~~ are only 2 Gates as opposed to LSTM 3 Gates. So naturally the trainable parameters are less and hence training is faster. In Weather Patterns where we don't require a long-term memory ~~to~~ and we can predict based on ~~psevi~~ some previous days, GRU can be a better option than LSTM.

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$$1.a \quad f(w) = \frac{1}{2} \|Xw - y\|^2$$



This is the graph for $|x|$ so at 0 it is not differentiable

$$f(x) = 1 \text{ for } x > 0$$

$$f(x) = -1 \text{ for } x < 0$$

$$\therefore \frac{d}{dx} f(w) = \frac{1}{2} \times 2x$$

Sr. No. 8499



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